

LLMs for AI

Comparison of LLMs for Rider Allocation , Container Allocation & Kabaddi Game



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Agenda

1. Problem Statement & Motivation
2. Rider Allocation Problem Analysis
3. Container Allocation Problem Analysis
4. Kabaddi Game Analysis
5. Overall Model Performance
6. Personal Experience
7. Prompts Reference



Motivation & Problem Statement



Problems Analyzed :

- Rider Allocation (Logistics optimization)
- Container Allocation (Packing optimization)
- Kabaddi Game (AI search algorithms)

Methodology: Comparative analysis using 4 difficulty levels per problem

LLMs Evaluated: GPT-4, Gemini, Claude

Models :

- **Claude Sonnet 4.5** (by Anthropic)
- **GPT-4.0** (by OpenAI)
- **Gemini 2.5 Pro** (by Google DeepMind)

Goal: Identify strengths/weaknesses for different AI problem types

THE RIDER ALLOCATION PROBLEM

Complex Logistical Puzzle

Uber Eats, DoorDash, Zomash, Deliveroo

KEY FACTORS IN THE ALLOCATION



Proximity



Rider's Current Location



Restaurant



Estimated Time of Delivery (ETD) & Efficiency



Traffic & Efficiency



Rider Availability & Status

THE PROBLEM

Factors often conflict

Nearest rider busy
Batching vs. Speed

THE GOAL

Minimize Delivery Time



Maximize Efficiency & Earnings

Ensure Fairness

Rider Allocation Problem

Rider Allocation - Easy Prompt

LLM Performance Summary:

- GPT-4: Correct solution (3 km). Concise, mathematical reasoning.
- Gemini: Correct solution (3 km). Clear, table-based explanation.
- Claude: Correct solution (3 km). Excelled by introducing advanced concepts like comparative advantage and the Hungarian Algorithm.

Key

All models solved this basic problem, but Claude demonstrated superior strategic and economic reasoning.

Takeaway:

PROBLEM: You are a dispatcher. You have 2 idle riders and 2 new orders.

- Rider A is 1 km from Restaurant X.
- Rider B is 3 km from Restaurant X.
- Rider A is 4 km from Restaurant Y.
- Rider B is 2 km from Restaurant Y.

What is the optimal assignment to minimize total travel distance to the restaurants? Explain your reasoning step-by-step. Also add a small gist and rationale for your answer.

Rider Allocation Problem

The Scenario



Optimal Solution

Assignment: Rider A → Restaurant X,
Rider B → Restaurant Y

Total Distance = 1 km + 2 km = 3 km

Gist

Assign riders to minimize combined travel. Rider A is closer to X, B is closer Y

Rationale

This assignment pairs each rider with their closet restaurant, reducing individual travel. Alternative: Rider A → Y (4km) = X (3km), Total = 7km. The optimal choice saves 4km.

Rider Allocation - Moderate Prompt

LLM Performance Summary:

GPT-4: Correctly identified the policy violation. Solid constraint checking.

Gemini: Correctly identified the policy violation. Direct and focused.

Claude: Correctly identified the violation. Excelled with a multi-factor analysis considering customer trust, risk management, and operational priorities.

Key Takeaway:

Claude again stood out by framing the problem as a strategic business decision, not just a rule-checking exercise.

PROBLEM: Rider Alex is 5 minutes from completing their first delivery. A new order appears. The restaurant is 2 minutes from Alex's current location, and the customer is 8 minutes from there. If assigned to Alex, this will delay his first delivery by 10 minutes. An idle rider, Sam, is 10 minutes from the new restaurant. The platform's policy is to avoid delaying existing orders by more than 8 minutes. The first customer's order is already 5 minutes behind the initial estimate. Should the new order be assigned to Alex or Sam? Justify your decision based on the constraints. Also add a small gist and rationale for your answer.

Rider Allocation Problem (Delay Policy)

The Scenario

Rider Alex



5 min to delivery
New order delay,
New order delay: 10 min
Policy limit: 8 min



POLICY
VIOLATION

Rider Sam



IDLE
10 min to Restaurant

Optimal Solution



Assign Order to Sam

Total Delay = 10 min (Sam) +
0 min, Alex) = 10 min

Gist

Assign new order to Sam to avoid violating delay policy for Alex's Alex's ongoing delivery.

Rationale

Alex is the closest but assigning the new to him violates delay policy. Sam further, but idle, assigning the only riders only thim is meet the policy requirement.

Rider Allocation - Difficult Prompt

LLM Performance Summary:

GPT-4: Strong quantitative analysis using a weighted scoring model.

Gemini: Basic priority-based reasoning.

Claude: Exceptional strategic depth. Integrated real-world storm context, risk management, and operational efficiency into the decision.

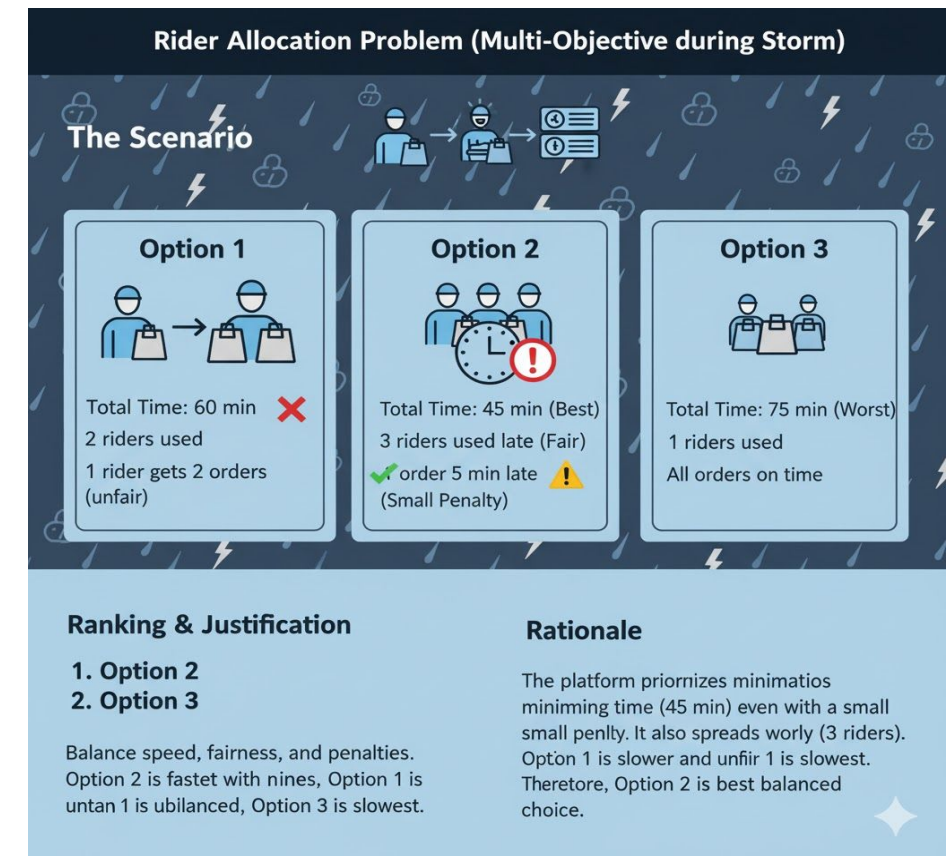
Key Takeaway:

For complex, multi-objective problems, Claude's ability to incorporate real-world context is unmatched.

PROBLEM: You have 3 riders and 3 high-priority orders during a storm. Assignment options:

- Option 1: Total delivery time is 60 min, uses 2 riders, but 1 rider gets 2 orders.
- Option 2: Total delivery time is 45 min, uses 3 riders, but one order is 5 min late (small penalty).
- Option 3: Total delivery time is 75 min, uses 1 rider, all orders on time. Also add a small gist and rationale for your answer.

The platform wants to: 1) Minimize total time, 2) Avoid penalties, 3) Spread work fairly. Rank these options from best to worst, justifying your ranking by balancing the competing objectives. Also add a small gist and rationale for your answer.



Rider Allocation - Very Difficult Prompt

LLM Performance Summary:

GPT-4: Correct calculation (20 min vs 30 min). Clear reasoning.

Gemini: Correct calculation (20 min vs 30 min). Direct and accurate.

Claude: Critical Failure. Made a fundamental math error (calculated 10 min vs 30 min), despite having the most sophisticated strategic framework.

Key Takeaway:

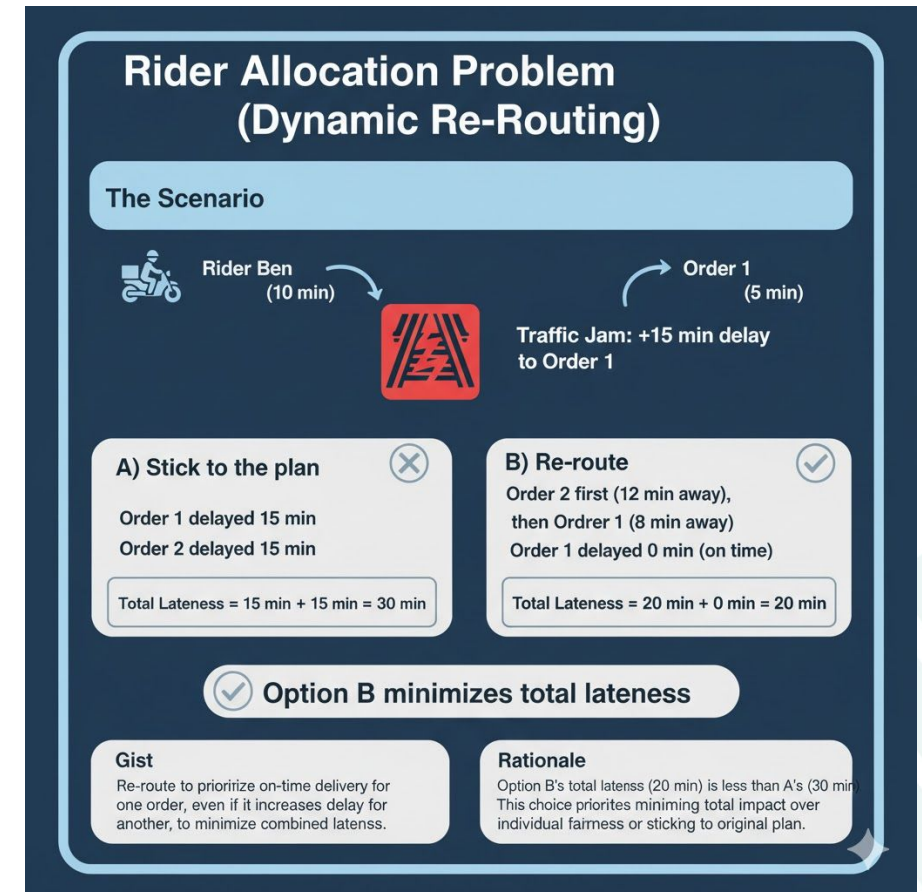
A critical flaw was revealed: Claude's advanced reasoning is undermined by occasional, severe calculation errors, making it risky for operational decisions

PROBLEM: Rider Ben is on a route to deliver Order 1 (10 min away) then Order 2 (another 5 min). A traffic jam will delay Order 1 by 15 minutes. You have two choices:

A) Stick to the plan: Order 1 delayed 15 min, Order 2 delayed 15 min.

B) Re-route: Ben delivers Order 2 first (now 12 min away), then Order 1 (another 8 min). This makes Order 1 20 min late but Order 2 on time.

Which option minimizes total lateness? Calculate the total lateness (sum of minutes late for all orders) for each option and choose the best one. Also add a small gist and rationale for your answer.



THE CONTAINER LOADING PROBLEM

Maximizing Volume in a Constrained Space

The Challenge



Pack a set of boxes (cuboids)



Into a single
Into a single container
(larger cuboid)



No overlap



The Goal



- ▶ Maximize Total Volume
- ▶ Maximize Value
- Efficient Packing Arrangement

Container Loading Problem

A shipping container has internal dimensions of 10 meters in length, 5 meters in width, and 4 meters in height. You have 40 identical boxes to pack. Each box is a cube with 2-meter sides.

Using the Container Loading Problem approach, what is the maximum number of these boxes that can be packed into the container? Explain how you would arrange them to achieve this number. Calculate both the total volume of the packed boxes and the utilization of the container's volume.

Container Loading - Easy Prompt

LLM Performance Summary:

GPT-4: Correct (20 boxes, 80% utilization). Clear step-by-step math.

Gemini: Correct (20 boxes, 80% utilization). Excellent explanation of physical vs. theoretical fit.

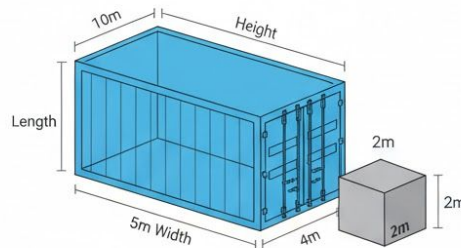
Claude: Critical Failure. Made a basic multiplication error ($5 \times 2 \times 2 = 10$ boxes).

Key Takeaway:

Claude's mathematical unreliability is a consistent weakness, even on simple geometric problems.

Container Allocation Problem (Single Box Type)

The Scenario



40 identical boxes, each 2m x 2m x 2m

Solving the Problem

1. Max boxes along each dimension

Length: $10\text{m} / 2\text{m} = 5$ boxes
Width: $5\text{m} / 2\text{m} = 2$ boxes

2. Total boxes

$5 \times 2 \times 2 = 20$ boxes

3. Arrangement:



Arrangement: 5 boxes long, 2 boxes wide, 2 boxes high.

Results

Total Volume of Packed Boxes
 $20 \text{ boxes} \times (2\text{m}^3/\text{box}) =$
 $20 \times 8 \text{ m}^3 = 160 \text{ m}^3$

Container Utilization

Container Volume:
Container Volume: $10\text{m} \times 5\text{m} \times 4\text{m} = 200 \text{ m}^3$
Utilization: $(160 \text{ m}^3 / 200 \text{ m}^3) \times 100\% = 80\%$

Maximum 20 boxes can be packed, achieving 80% utilization

Gist

Divide container dimensions by box size to find how many those numbers for total.

Rationale

The dimensions only allow for 2 high s and 2 boxes wide, long, limiting the total.

Consider a container with dimensions 12 ft (L) x 8 ft (W) x 10 ft (H). You have three types of boxes to load:

Type A: 20 boxes, each 4 ft x 4 ft x 4 ft.

Type B: 15 boxes, each 6 ft x 4 ft x 2 ft.

Type C: 10 boxes, each 3 ft x 3 ft x 3 ft.

All boxes can be rotated. Your goal is to maximize the total number of boxes loaded. Propose a high-level packing strategy. Which boxes will you load first and in what orientation? What is the theoretical maximum number of boxes you could fit if volume was the only constraint, and why might this not be achievable in practice?

Container Allocation - Medium Prompt

LLM Performance Summary:

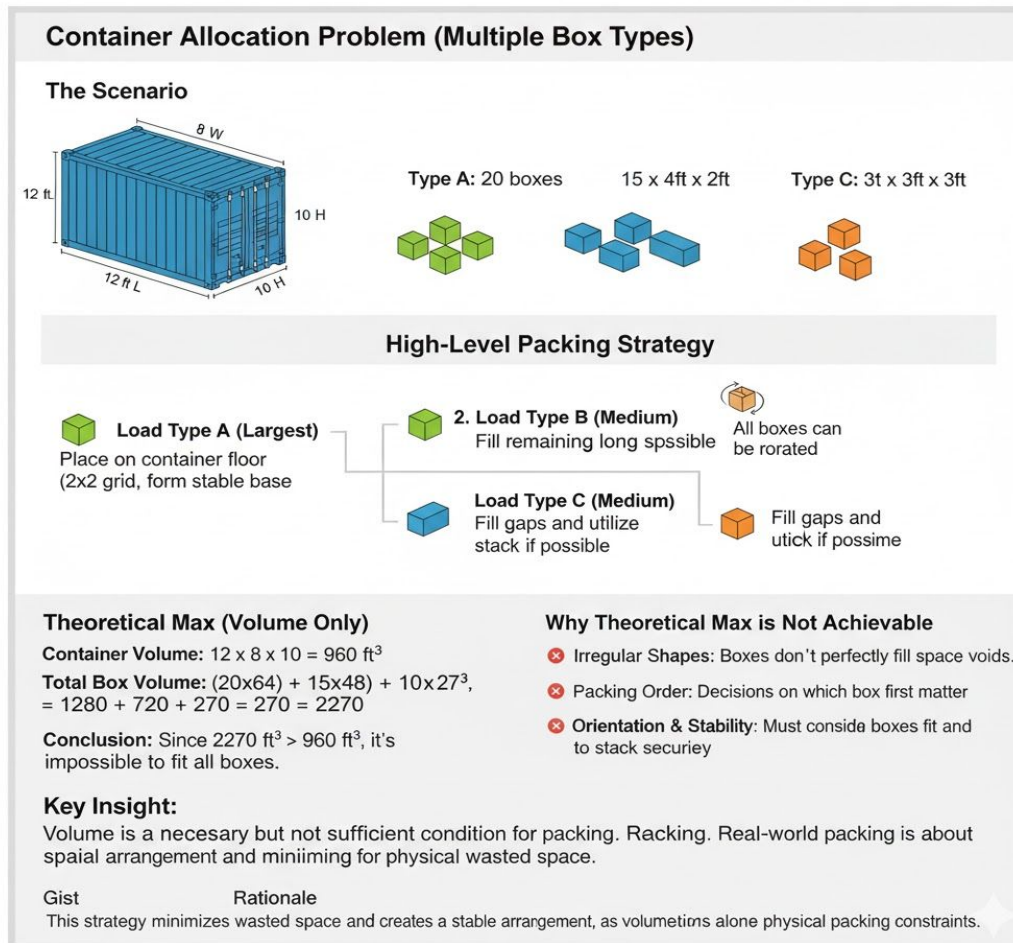
GPT-4: Correct theoretical maximum. Good "small-first" strategy.

Claude: Incorrect theoretical maximum. Confused strategy.

Gemini: Standout Performer. Correct calculations and demonstrated the optimal "largest-first" strategy for creating stable, clean subspaces.

Key Takeaway:

Gemini excelled in pedagogical value and demonstrated superior fundamental packing logic.



Container Allocation - Hard Prompt

You are loading a 20-foot container (internal dimensions: 19.5 ft L x 7.5 ft W x 7.5 ft H). You must load the following items:

Pallets: 5 heavy, non-stackable metal pallets. Each is 4 ft x 4 ft x 0.5 ft and weighs 200 lbs.

Crates: 12 sturdy wooden crates. Each is 3 ft x 2 ft x 2 ft and can be stacked.

Boxes: 25 fragile cardboard boxes. Each is 2 ft x 1.5 ft x 1 ft. They can be stacked, but only on top of each other (nothing heavier can be placed on them).

Constraints:

The total weight must be evenly distributed; the heavy pallets cannot all be placed on one side.

The fragile boxes must be loaded last and placed on top of the stack to prevent crushing.

The container must be stable for transport; a lower center of gravity is better.

Devise a loading plan. Describe the sequence of loading (what goes in first, what goes in last) and the general arrangement of items to satisfy all constraints. Do not provide exact coordinates, but describe the logic of your packing strategy.

LLM Performance Summary:

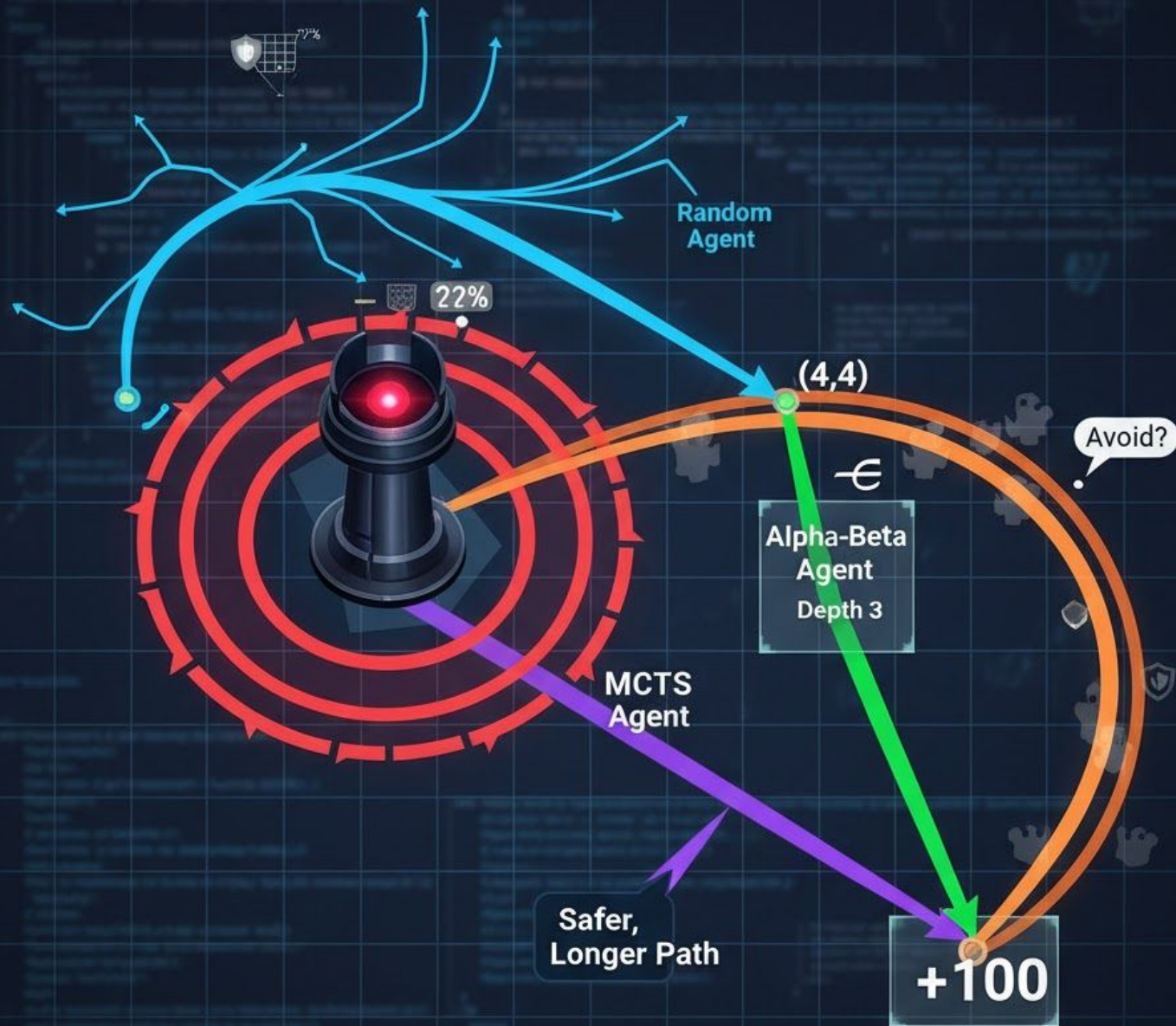
Gemini: Basic but correct three-phase approach.

Claude: Good structure but with impractical suggestions.

GPT-4: Exceptional Performance. Provided a comprehensive, multi-phase plan with operational expertise, detailed weight distribution, and safety considerations.

Key Takeaway:

For complex, real-world constraint satisfaction, GPT-4 is the clear leader in practicality and depth.



Kabaddi Game

Kabaddi Game - Easy Instance



LLM Performance Summary:

Gemini: Provided structured theoretical predictions.

Claude: Excellent theoretical depth with detailed tactical scenarios.

GPT-4: Provided unique empirical data from actual simulations, offering transparency about parameters and limitations.

Key Takeaway:

Claude for theory, GPT-4 for practice. GPT-4's ability to run and report on simulations is a significant advantage.

Problem Setup:

Board: 4x4 grid (Team A territory: columns 0-1, Team B territory: columns 2-3)

Team A Start: (0,0) and (1,0)

Team B Start: (2,3) and (3,3)

Treasure A: (0,1)

Treasure B: (3,2)

Game Type: Turn-based

Task:

Using the Kabaddi game environment with 4 AI agents (Random, Greedy, Alpha-Beta, MCTS), simulate this scenario for 50 games each and answer:

Which agent type wins most frequently as Team A? As Team B?

What is the average game length (steps) for each agent pairing?

Which agent makes the most strategic captures?

Expected Analysis:

Greedy agents should perform well due to direct paths to treasures

Random agents should have poor performance

Compare how search depth affects Alpha-Beta vs MCTS in this simple space

Kabaddi Game - Moderate & Hard Instances



LLM Performance Summary:

GPT-4: Strong practical guidance for experimental setup.

Gemini: Basic conceptual understanding.

Claude: Exceptional analytical performance. Provided quantitative predictions, multi-phase strategies, and a sophisticated analysis of algorithm scalability, identifying the precise "crossover point" where MCTS outperforms Alpha-Beta.

Key Takeaway:

Claude demonstrates expert-level understanding of game AI principles and search algorithm behavior.

Moderate Instance:

Board: 6x6 grid (Team A: columns 0-2, Team B: columns 3-5)

Team A Start: (0,0) and (2,2)

Team B Start: (3,5) and (5,3)

Treasure A: (1,1) - guarded position

Treasure B: (4,4) - central location

Game Type: Simultaneous moves

Hard Instance:

Board: 8x8 grid (Team A: columns 0-3, Team B: columns 4-7)

Team A Start: (0,0) and (3,2)

Team B Start: (7,7) and (4,5)

Treasure A: (2,3) - behind defensive line

Treasure B: (5,4) - neutral zone edge

Special Rule: Players get +1 movement when carrying treasure

Game Type: Simultaneous moves with advanced capture rules

Tasks for Moderate & Hard:

Run 100 simulations for each agent pairing across both instances

Analyze:

Win rates for each agent type in both offensive and defensive roles

Capture-to-death ratios for strategic evaluation

How game complexity affects Alpha-Beta vs MCTS performance

Whether simultaneous movement changes optimal strategies

Key Questions:

Does MCTS outperform Alpha-Beta as state space grows?

How do greedy agents fail in complex tactical situations?

What's the trade-off between treasure hunting and player capture strategies?

Which agent best balances offense and defense?

Overall Analysis - Best Model by Case



Rider Allocation:

Claude (Strategy) / GPT-4 (Execution)

Container Loading:

GPT-4 (Complex) / Gemini (Education)

Kabaddi Game:

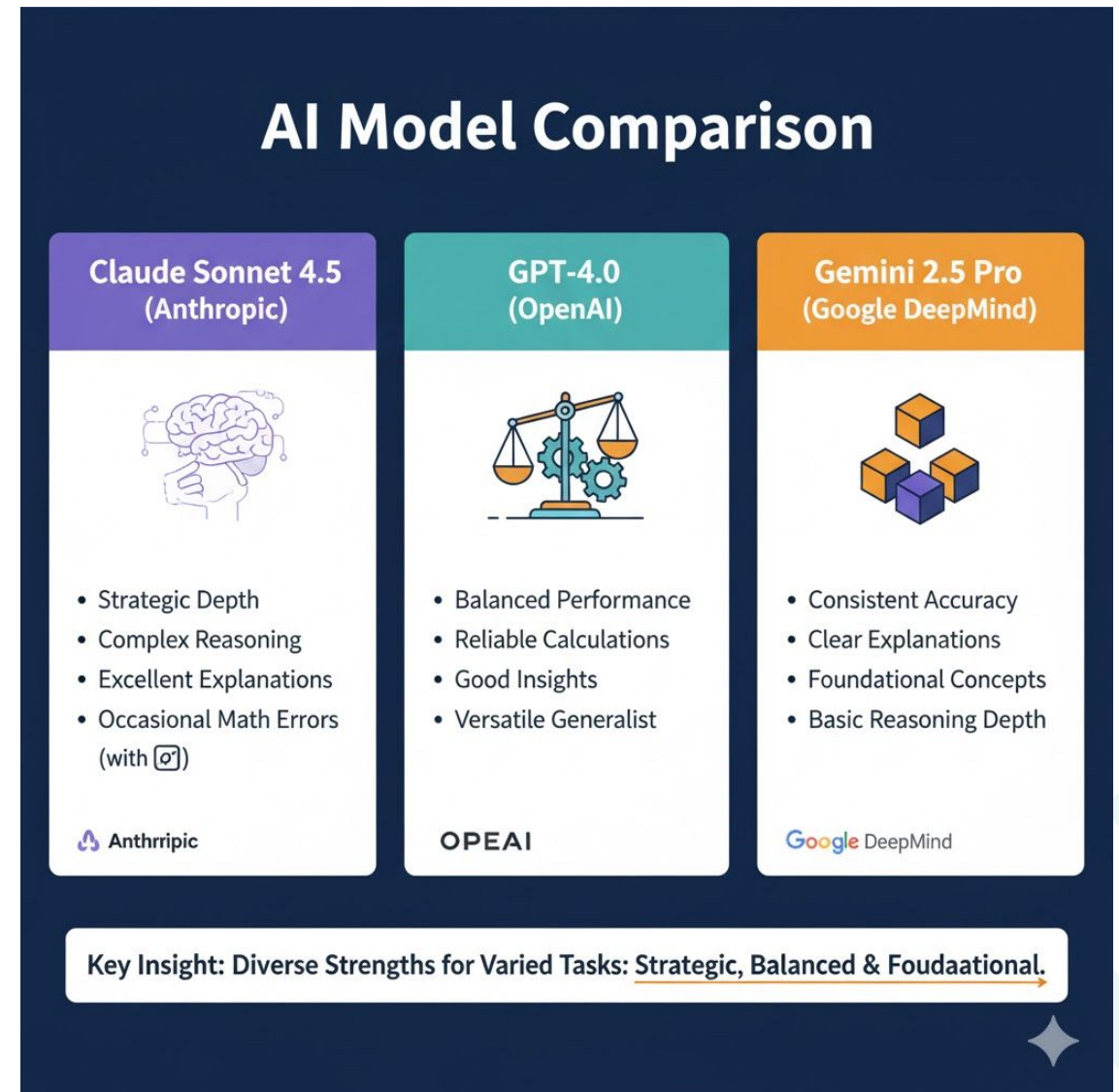
Claude (Theory) / GPT-4 (Simulation)

Mathematical Reliability:

GPT-4 > Gemini > Claude

Strategic Depth:

Claude > GPT-4 > Gemini



Personal Experience & Convenience



- GPT-4: Most reliable for production use
- Claude: Best for brainstorming and strategic planning
- Gemini: Excellent for quick explanations and teaching
- Recommendation: Use Claude + GPT-4 combination with cross-verification



References



Rider Allocation

[GPT-4.0](#) • [Claude Sonnet 4.5](#)

Container Loading

[GPT-4.0](#) • [Claude Sonnet 4.5](#)

Kabaddi Game

[GPT-4.0](#) • [Claude Sonnet 4.5](#)

Submission Drive

https://drive.google.com/drive/folders/1IMFnd0kQGM6UBqczvH1er9zmeT_IWs24?usp=sharing

! Thank You !